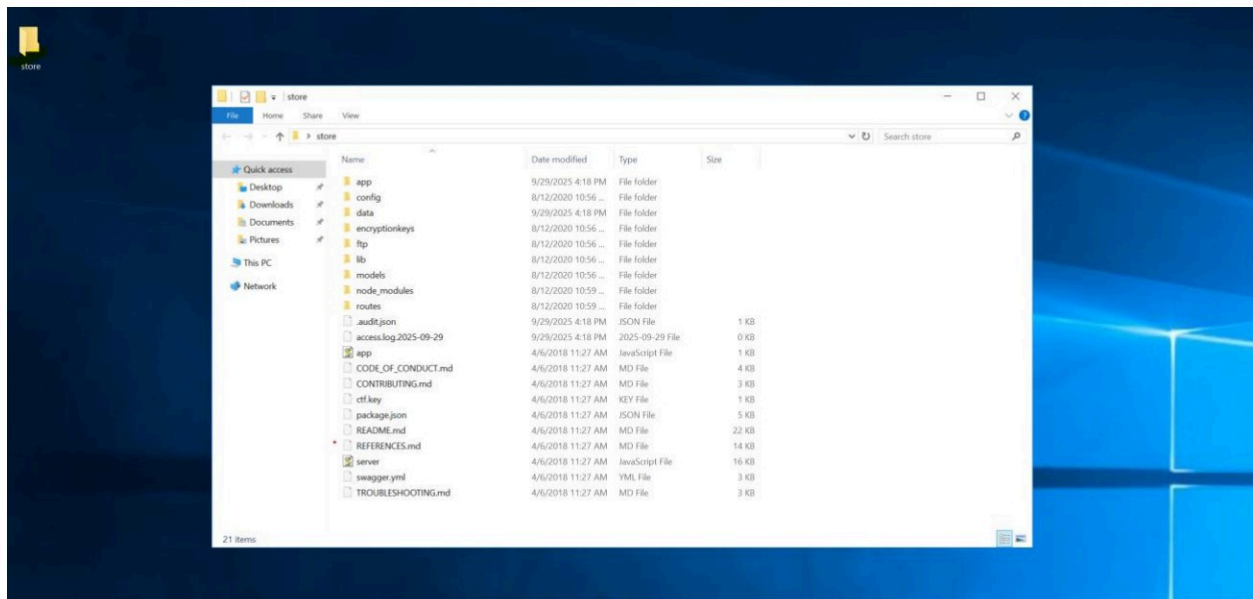


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NMAP Lab

To start off, I opened the Windows Server and then navigated to the “store” folder on the desktop



From here, I simply clicked and then opened a command window from the directory and entered the “npm start” command. As you can see in the yellow text below, the web server hosted from this VM is listening on port 80, the standard port for HTTP.

```
npm
C:\Users\Administrator\Desktop\store>npm start

> juice-shop@7.0.2 start C:\Users\Administrator\Desktop\store
> node app

2025-09-29T20:18:37.247Z '[FileStreamRotator] Could not remove old log file: ' './access.log.2020-08-12'
Server listening on port 80
```

Now that we have verified the Windows server is running as intended, we can switch back over to our Kali VM. Once on our Kali VM, I opened the terminal and entered “msfconsole” to start the Metasploit service.

```
(kali@kali)-[~]
$ msfconsole
Metasploit tip: Writing a custom module? After editing your module, why not try
the reload command

Metasploit

      =[ metasploit v6.4.64-dev ]
+ -- --=[ 2519 exploits - 1296 auxiliary - 431 post ]
+ -- --=[ 1607 payloads - 49 encoders - 13 nops ]
+ -- --=[ 9 evasion ]

Metasploit Documentation: https://docs.metasploit.com/

msf6 > 
```

To perform a basic scan with nmap, we can use the “sn” command to do so. As you can see, the Windows VM is present on the scan, the address being **REDACTED**.

```
Nmap done: 1 IP address (1 host up) scanned in 14.58 seconds
msf6 > nmap -sn REDACTED
[*] exec: nmap -sn REDACTED

Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-29 16:52 EDT
Nmap scan report for REDACTED
Host is up (0.00019s latency).
MAC Address: REDACTED (VMware)
Nmap scan report for REDACTED
Host is up (0.00057s latency).
MAC Address: REDACTED (VMware)
Nmap scan report for REDACTED
Host is up (0.0014s latency).
MAC Address: REDACTED (VMware)
Nmap scan report for REDACTED
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 27.92 seconds
msf6 > 
```

To find out more information about the Windows system specifically, we can simply do “nmap **REDACTED**”

As pictured below, we were able to gather data about the system such as which ports and services it is using, and the status of them.

```
[*] exec: nmap REDACTED
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-29 16:53 EDT
Nmap scan report for REDACTED
Host is up (0.00038s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
443/tcp   open  https
5985/tcp   open  wsman
MAC Address: REDACTED (VMware)

Nmap done: 1 IP address (1 host up) scanned in 17.92 seconds
```

To perform an even more advanced scan, we can enter the command “nmap -sS -Pn -A **REDACTED**”. This will show us even more information about the network that a basic “sn” scan will not be able to do.

REDACTED

To scan a specific port, we can enter the command “nmap -p 80 REDACTED”

(I finished the lab from my personal PC at home, which is why the network address/ VM addresses are now different)

From the results below, we can see that the address “REDACTED” (Windows Server) uses port 80 and the http service that coincides with it.

```
[*] exec: nmap -p 80 REDACTED
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-29 20:40 EDT
Nmap scan report for REDACTED
Host is up (0.00015s latency).

PORT      STATE SERVICE
80/tcp    closed http
MAC Address: REDACTED (VMware)

Nmap scan report for REDACTED
Host is up (0.00046s latency).

PORT      STATE SERVICE
80/tcp    filtered http
MAC Address: REDACTED (VMware)

Nmap scan report for REDACTED
Host is up (0.000076s latency).

PORT      STATE SERVICE
80/tcp    filtered http
MAC Address: REDACTED (VMware)

Nmap scan report for REDACTED
Host is up (0.000023s latency).

PORT      STATE SERVICE
80/tcp    closed http

Nmap done: 256 IP addresses (4 hosts up) scanned in 28.25 seconds
msf6 > 
```

Continuing upon this, we can enter the command “nmap -sU -p 53,161,162 REDACTED” to scan the network for three ports associated with UDP (53,161, and 162).

```
[*] exec: nmap -sU -p 53,161,162 REDACTED

Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-29 20:45 EDT
Nmap scan report for REDACTED
Host is up (0.000074s latency).

PORT      STATE      SERVICE
53/udp    closed    domain
161/udp    closed    snmp
162/udp    closed    snmptrap
MAC Address: 00:50:56:C0:00:01 (VMware)

Nmap scan report for REDACTED
Host is up (0.00020s latency).

PORT      STATE      SERVICE
53/udp    open|filtered domain
161/udp    open|filtered snmp
162/udp    open|filtered snmptrap
MAC Address: 00:0C:29:94:2E:9A (VMware)

Nmap scan report for REDACTED
Host is up (0.000063s latency).

PORT      STATE      SERVICE
53/udp    open|filtered domain
161/udp    open|filtered snmp
162/udp    open|filtered snmptrap
MAC Address: 00:50:56:E8:91:28 (VMware)

Nmap scan report for REDACTED
Host is up (0.000037s latency).

PORT      STATE      SERVICE
53/udp    closed    domain
161/udp    closed    snmp
162/udp    closed    snmptrap
```

We can also try “r nmap -sV --version-intensity 9 REDACTED” to perform a service scan of the Windows server. The results of this scan revealed some information such as the OS and the fact that it currently is using Microsoft IIS on port 443, or the default port for HTTPS.


```
[*] exec: nmap -sV --version-intensity 9 REDACTED

Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-29 20:48 EDT
Nmap scan report for REDACTED
Host is up (0.00030s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
443/tcp    open  ssl/http Microsoft IIS httpd 10.0
5985/tcp    open  http Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
MAC Address: 00:0C:29:94:2E:9A (VMware)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 30.03 seconds
msf6 > 
```

To find more potential information while doing active reconnaissance, we can use Nmap scripts that are readily available to use in Kali Linux. To do this, simply type “locate .nse | grep script.”

```
[*] exec: locate .nse | grep script

/usr/share/legion/scripts/nmap/shodan-api.nse
/usr/share/legion/scripts/nmap/shodan-hq.nse
/usr/share/legion/scripts/nmap/vulners.nse
/usr/share/nmap/scripts/acarsd-info.nse
/usr/share/nmap/scripts/address-info.nse
/usr/share/nmap/scripts/afp-brute.nse
/usr/share/nmap/scripts/afp-ls.nse
/usr/share/nmap/scripts/afp-path-vuln.nse
/usr/share/nmap/scripts/afp-serverinfo.nse
/usr/share/nmap/scripts/afp-showmount.nse
/usr/share/nmap/scripts/ajp-auth.nse
/usr/share/nmap/scripts/ajp-brute.nse
/usr/share/nmap/scripts/ajp-headers.nse
/usr/share/nmap/scripts/ajp-methods.nse
/usr/share/nmap/scripts/ajp-request.nse
/usr/share/nmap/scripts/allseeingeeye-info.nse
/usr/share/nmap/scripts/amqp-info.nse
/usr/share/nmap/scripts/asn-query.nse
/usr/share/nmap/scripts/auth-owners.nse
/usr/share/nmap/scripts/auth-spoof.nse
/usr/share/nmap/scripts/backorifice-brute.nse
/usr/share/nmap/scripts/backorifice-info.nse
/usr/share/nmap/scripts/bacnet-info.nse
/usr/share/nmap/scripts/banner.nse
/usr/share/nmap/scripts/bitcoin-getaddr.nse
/usr/share/nmap/scripts/bitcoin-info.nse
/usr/share/nmap/scripts/bitcoinrpc-info.nse
/usr/share/nmap/scripts/bittorrent-discovery.nse
```

To have these scripts readily available, I entered command “locate .nse | grep script > nsescripts.txt” to pipe the previous results into a text file.

```
[*] exec: locate .nse | grep script > nsescripts.txt
```

To perform a simple service scan of our server without using any specific scripts, we can use the command “nmap -sV -sC REDACTED” to view information such as the ssl-cert validity.

```
[*] exec: nmap -sV -sC REDACTED

Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-29 20:55 EDT
Nmap scan report for REDACTED
Host is up (0.00028s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
443/tcp   open  ssl/http Microsoft IIS httpd 10.0
|_http-title: IIS Windows Server
|_ssl-date: 2025-09-30T00:55:46+00:00; +1s from scanner time.
|_ssl-cert: Subject: commonName=Server00
|_Not valid before: 2020-08-11T17:52:59
|_Not valid after: 2021-08-11T00:00:00
|_tls-alpn:
|_  h2
|_  http/1.1
|_http-methods:
|_  Potentially risky methods: TRACE
|_http-server-header: Microsoft-IIS/10.0
5985/tcp  open  http      Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
|_http-title: Not Found
MAC Address: 00:0C:29:94:2E:9A (VMware)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 34.64 seconds
```

Now using the .nse scripts, we will attempt to see if the Windows server is vulnerable to the “heartbleed” vulnerability.

To get a description of the heartbleed vulnerability, we can use command “nmap --script-help=ssl-heartbleed”.

```
[*] exec: nmap --script-help=ssl-heartbleed

Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-29 20:58 EDT

ssl-heartbleed
Categories: vuln safe
https://nmap.org/nsedoc/scripts/ssl-heartbleed.html
  Detects whether a server is vulnerable to the OpenSSL Heartbleed bug (CVE-2014-0160).
  The code is based on the Python script ssltest.py authored by Katie Stafford (katie@ktpanda.org)
msf6 >
```

Now to use the script in an actual scan, I entered the command “nmap -sV REDACTED --script=ssl-heartbleed”.

As you can see below, the results are that of a typical service scan. This means the Windows server is not vulnerable to the heartbleed vulnerability. Had this not been the case, the results would have indicated this with something like “VULNERABLE” appearing.

```
[*] exec: nmap -sV REDACTED --script=ssl-heartbleed

Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-29 21:01 EDT
Nmap scan report for REDACTED
Host is up (0.00027s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
443/tcp    open  ssl/http     Microsoft IIS httpd 10.0
|_http-server-header: Microsoft-IIS/10.0
5985/tcp    open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
MAC Address: REDACTED (VMware)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 29.68 seconds
```

Performing Nmap scans for ports can reveal crucial information about a system, subnet, or network.

To perform a scan for just the first 100 ports, we could use the command “nmap -p 1-100 REDACTED”

```
[*] exec: nmap -p 1-100 192.168.23.0/24

Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-29 21:04 EDT
Nmap scan report for REDACTED
Host is up (0.000060s latency).
All 100 scanned ports on REDACTED are in ignored states.
Not shown: 100 closed tcp ports (reset)
MAC Address: REDACTED (VMware)

Nmap scan report for REDACTED
Host is up (0.00015s latency).
All 100 scanned ports on REDACTED are in ignored states.
Not shown: 100 filtered tcp ports (no-response)
MAC Address: REDACTED (VMware)

Nmap scan report for REDACTED
Host is up (0.000066s latency).
All 100 scanned ports on REDACTED are in ignored states.
Not shown: 100 filtered tcp ports (no-response)
MAC Address: REDACTED (VMware)

Nmap scan report for REDACTED
Host is up (0.000040s latency).
All 100 scanned ports on REDACTED are in ignored states.
Not shown: 100 closed tcp ports (reset)

Nmap done: 256 IP addresses (4 hosts up) scanned in 30.15 seconds
```

To perform a scan of the 100 most used ports, we could enter the command “nmap -F REDACTED”.


```
[*] exec: nmap -F 1 REDACTED

Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-29 21:05 EDT
Nmap scan report for REDACTED
Host is up (0.10s latency).
Not shown: 96 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
5357/tcp   open  wsdapi
MAC Address: REDACTED (VMware)

Nmap scan report for REDACTED
Host is up (0.00013s latency).
Not shown: 99 filtered tcp ports (no-response)
PORT      STATE SERVICE
443/tcp    open  https
MAC Address: REDACTED (VMware)

Nmap scan report for REDACTED
Host is up (0.000097s latency).
All 100 scanned ports on REDACTED are in ignored states.
Not shown: 100 filtered tcp ports (no-response)
MAC Address: REDACTED (VMware)

Nmap scan report for REDACTED
Host is up (0.0000040s latency).
All 100 scanned ports on REDACTED are in ignored states.
Not shown: 100 closed tcp ports (reset)

Nmap done: 256 IP addresses (4 hosts up) scanned in 30.08 seconds
```

To perform a scan over all 65,535 ports over the subnet, enter the command “nmap -p-
REDACTED”.

```

[*] exec: nmap -p- REDACTED
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-29 21:17 EDT
Nmap scan report for REDACTED
Host is up (0.00011s latency).
Not shown: 65518 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
902/tcp    open  iss-realsure
912/tcp    open  apex-mesh
1462/tcp   open  world-lm
5040/tcp   open  unknown
5357/tcp   open  wsddapi
7680/tcp   open  pando-pub
49664/tcp  open  unknown
49665/tcp  open  unknown
49666/tcp  open  unknown
49667/tcp  open  unknown
49668/tcp  open  unknown
49669/tcp  open  unknown
57239/tcp  open  unknown
57621/tcp  open  unknown
MAC Address: REDACTED (VMware)

Nmap scan report for REDACTED
Host is up (0.00034s latency).
Not shown: 65533 filtered tcp ports (no-response)
PORT      STATE SERVICE
443/tcp    open  https
5985/tcp   open  wsman
MAC Address: REDACTED (VMware)

Nmap scan report for REDACTED
Host is up (0.000070s latency).
All 65535 scanned ports on REDACTED are in ignored states.
Not shown: 65535 filtered tcp ports (no-response)
MAC Address: REDACTED (VMware)

Nmap scan report for REDACTED
Host is up (0.000040s latency).
All 65535 scanned ports on REDACTED are in ignored states.
Not shown: 65535 closed tcp ports (reset)

Nmap done: 256 IP addresses (4 hosts up) scanned in 416.31 seconds
msf6 >

```

If we wanted to perform a scan using only specific IP addresses, we can do so by creating a text file and telling Nmap to scan the addresses found in the file.

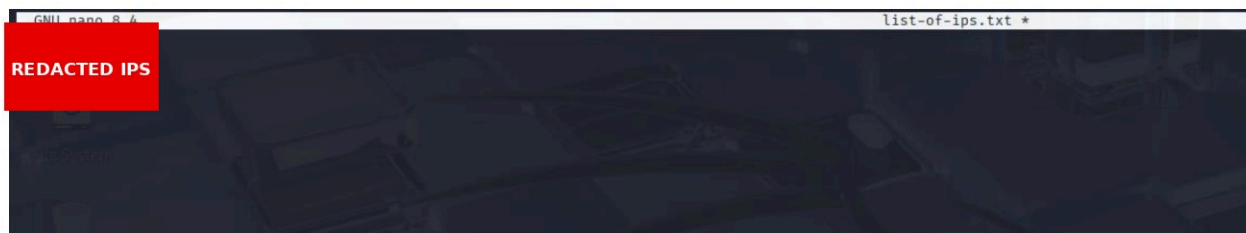
To start, I created a file called “list-of-ips.txt” in the home directory.

```

[*] exec: touch list-of-ips.txt
msf6 > ls
[*] exec: ls
Desktop Documents Downloads http_mcdonalds.com_robots.txt list-of-ips.txt mcdonaldsreport.html Music nsescripts.txt Pictures Public Templates Videos

```

In this file, I entered the four addresses of the systems that were present in my previous Nmap scans.



After saving the text file, in the Metasploit console line I entered command “Nmap -iL list-of-ips.txt.”

As pictured below, all the IP’s listed in the text file were scanned.

```
[*] exec: nmap -iL list-of-ips.txt

Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-29 21:12 EDT
Nmap scan report for REDACTED
Host is up (0.00020s latency).
Not shown: 994 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
902/tcp    open  iss-realsecure
912/tcp    open  apex-mesh
5357/tcp   open  wsdapi
MAC Address: REDACTED (VMware)

Nmap scan report for REDACTED
Host is up (0.00026s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
443/tcp    open  https
5985/tcp   open  wsman
MAC Address: REDACTED (VMware)

Nmap scan report for REDACTED
Host is up (0.000087s latency).
All 1000 scanned ports on REDACTED are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: REDACTED (VMware)

Nmap scan report for REDACTED
Host is up (0.000040s latency).
All 1000 scanned ports on REDACTED are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 4 IP addresses (4 hosts up) scanned in 37.72 seconds
```

To export the results of this scan to a text file, we can use the command “Nmap -iL list-of-ips.txt -oN myfile.txt.” This will output the results of the scan from “list-of-ips.txt” to a file called “myfile.txt”.

As pictured below, you can see the contents of “myfile.txt” match that of the scan performed on the text file.

```
GNU nano 8.4 myfile.txt
# Nmap 7.95 scan initiated Mon Sep 29 21:14:27 2025 as: /usr/lib/nmap/nmap --privileged -iL list-of-ips.txt -oN myfile.txt
Nmap scan report for REDACTED
Host is up (0.000075s latency).
Not shown: 994 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
902/tcp    open  iss-realsure
912/tcp    open  apex-mesh
5357/tcp   open  wsddapi
MAC Address: REDACTED (VMware)

Nmap scan report for REDACTED
Host is up (0.00027s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
443/tcp    open  https
5985/tcp   open  wsman
MAC Address: REDACTED (VMware)

Nmap scan report for REDACTED
Host is up (0.00011s latency).
All 1000 scanned ports on REDACTED are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: REDACTED (VMware)

Nmap scan report for REDACTED
Host is up (0.0000040s latency).
All 1000 scanned ports on REDACTED are in ignored states.
Not shown: 1000 closed tcp ports (reset)

# Nmap done at Mon Sep 29 21:15:03 2025 -- 4 IP addresses (4 hosts up) scanned in 36.55 seconds
```